Understanding the Soil Bearing Resistance in a Different Gravity Environment via Particle Density Scaling

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ABSTRACT

Understanding the bearing resistance of planetary regolith is essential to establish a long-term presence on the planets, e.g., extraterrestrial human habitat construction. Despite its importance, our understanding of the bearing resistance remains at best incomplete due to the dearth of data collected from the planetary surface missions as well as the limited amount of planetary soil samples available for the study. Moreover, it is hard to experimentally test the bearing resistance in a different gravity condition that can be obtained through parabolic flight, but which is expensive and can simulate a relatively short duration of different gravities. On the other hand, numerical studies using the discrete element method typically require high computational cost. This study hypothesizes a new approach to investigate the soil bearing resistance in a different gravity environment by simply scaling the particle density instead of reproducing the different gravity conditions. To this end, Terzaghi's bearing capacity theory is tweaked and used to develop the hypothesis. This study performs a set of discrete element simulations to test the hypothesis and demonstrate the feasibility of the proposed approach.

INTRODUCTION

Extraterrestrial soil research is at the core of planetary missions to understand the geomechanical properties including the soil bearing resistance against surface loading. This effort is vital to enable successful future surface activities such as the human habitat construction (Lim et al. 2017). In spite of the importance of the space mission-related geomechanics study, our understanding is still limited due to the accessibility issues. Seven Surveyor space crafts have been sent from 1966 to 1968 to investigate the geotechnical properties of the lunar surface. Two of them (Surveyor III and VII) were loaded with the soil mechanics surface sampler for various in-situ testing on the lunar surface (Scott and Roberson 1969). However, the data obtained from the in-situ testing is not sufficient to understand the soil bearing resistance. A limited quantity of lunar soil was brought back to Earth, but which was not enough for experiments needed to understand the geomechanical behavior. Therefore, a number of simulants including Johnson Space Center lunar soil simulant (JSC-1A) have been developed for use in experiments (Zeng et al. 2010). The mechanical properties including the soil bearing resistance of JSC-1A were tested through a series of laboratory experiments (Bucek et al. 2008) but the effect of gravity was not considered due to the challenge of simulating different gravities in a laboratory set up. For this reason, experiment using parabolic flight has been adopted in the research community to simulate the effect of different gravity. Kobayashi et al. (2009) and Marshall et al. (2018) conducted the partial gravity experiment using the parabolic flight to investigate the gravity effect on the

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soil bearing resistance. However, the experiments that could be performed during the flight were limited due to the relatively short duration of partial gravity conditions that could be simulated up to a maximum of 20-40 seconds. Numerical simulation using, e.g., discrete element method (DEM), enables to obtain high fidelity results while varying the gravitational acceleration. Lee et al. (2010) showed the effectiveness of DEM simulation to understand the soil bearing resistance using polyhedral particles that present realistic particle shapes. However, such numerical approach suffered from the high computational costs, which kept researchers from performing multiple simulations for systematic parametric studies.

To bypass the existing constraints, this study hypothesizes a new approach to investigate the soil bearing resistance in a different gravity environment by simply scaling the particle density instead of reproducing the different gravity conditions, e.g., through expensive parabolic flight. The theoretical background of this study builds upon Terzaghi's bearing capacity theory, which is discussed further in the following section.

Terzaghi's Bearing Capacity Theory

Terzaghi's formula to estimate the soil bearing capacity is shown in Equation (1) which shows the contribution by cohesion, surcharge, and weight of soil (Terzaghi et al. 1996).

$$q_d = cN_c + \gamma D_f N_q + 0.5\gamma B N_\gamma = cN_c + \gamma\left(D_f N_q + 0.5BN_\gamma\right) \tag{1}$$

where

q_d : bearing capacity of a soil mass located above water table

c : cohesion intercept

γ : unit weight of soil

D_f, B: depth and width of footing

N_c, N_q, N_γ : bearing capacity factors with respect to cohesion, surcharge, and weight of soil

The bearing capacity factors (N_c, N_q, N_γ) are dimensionless quantities that depend on soil friction angle ϕ'. Therefore, the three soil properties that contribute to the bearing resistance are c, γ and ϕ'. Considering the cohesion of lunar soil is relatively low (Kobayashi et al. 2009; Mitchell et al. 1972; Zeng et al. 2010), the effect of the first term (cN_c) is negligible, which leaves ϕ' and γ as the controlling parameters. It was reported that ϕ' does not depend on gravity (Marshall et al. 2018; Zou et al. 2015). However, γ has the gravity dependence because $\gamma = \rho g$ where ρ is the soil density and g is gravitational acceleration. Equation (1) can be presented as Equation (2) for a partial gravity condition (expressed as g/n) which indicates Terzaghi's formula can be adopted to estimate the soil bearing resistance in a different gravity condition.

$$q_d = \rho(g/n)(D_f N_q + 0.5BN_\gamma) \tag{2}$$

Equation (2) may be presented as Equation (3), where a partial density ρ/n is considered in Earth gravity g. Therefore, we can hypothesize from Equations (2) and (3) that *if Terzaghi's formula holds true for a different gravity environment, then the soil bearing resistance of a planet may be reproduced in the Earth gravity condition by proportionally scaling the particle density*.

$$q_d = (\rho/n)g(D_f N_q + 0.5BN_\gamma) \tag{3}$$

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Objective and Overview

This study tweaks the Terzaghi's formula and aims to evidence whether the soil bearing resistance in a different gravity environment can be reproduced by simply scaling the particle density, for which discrete element analysis is performed. Water is not considered in this study. The parabolic flight experiments performed by Kobayashi et al. (2009) in two reduced gravity conditions ($1/2g$, $1/6g$) are first reproduced by the numerical simulation. The density of modeled particles is then scaled to $1/2\rho$ and $1/6\rho$ and simulated in Earth gravity (g). The results are then compared.

Expected Significance

With the hypothesis successfully tested, this study will allow for use of 3D-printed synthetic particles with scaled density in a general laboratory set up to understand the extraterrestrial soil behavior. For example, the density of 3D-printed object from a typical photopolymer resin is ~1 g/cm^3 (Formlabs 2014), which is about 38% of typical mineral grains with 2.6 g/cm^3 of density. Considering the gravitational acceleration of Mars is 3.7 m/s^2 that is about 38% of the gravitational acceleration of Earth, the 3D-printed particles can be used to experimentally simulate the bearing resistance of Martian soil in the Earth gravity condition. Watters and Bernhardt (2018) and Zguris (2016) reported the compressive strength and elastic modulus of the photopolymer-based 3D-printed objects, which are roughly comparable to those of Florida limestone (Su et al. 2020), i.e., reasonably hard and strong enough. This study will not only introduce a new approach to the space mission-related research and but also improve students' interest and understanding of the planetary exploration.

PARABOLIC FLIGHT EXPERIMENT ON SOIL BEARING RESISTANCE IN REDUCED GRAVITY CONDITIONS

Kobayashi et al. (2009) performed parabolic flight experiment to observe the soil bearing resistance for a set of different gravity conditions. Two types of soils were considered: (i) uniformly graded Toyoura sand with D_{50} and D_{10} that are 0.26 and 0.21 mm; and (ii) well-graded Japanese lunar soil simulant (FJS-1) whose D_{50} and D_{10} are 0.10 and 0.014 mm. The soil was heaped up in a rectangular box whose interior area was 400 mm×50 mm and 160 mm in height. The model footing with base area of 20 mm×50 mm was located above the soil bed and vertically pushed during the flight, for which a loading rate of 3 mm/sec was considered. The experiment data was obtained for different gravity levels: $0g$, $1/6g$, $1/2g$, $3/4g$, and $1g$. Each gravity was maintained for 20-40 seconds. Interested readers are directed to Kobayashi et al. (2009) for more details.

DISCRETE ELEMENT MODELING

An open source DEM code, LIGGGHTS (DCS Computing Gmbh 2016a), is used for numerical study. Among the two types of soils used in the parabolic flight experiment, Toyoura sand with 60% relative density (void ratio ~ 0.76) is selected for the numerical modeling. The measured size distribution of Toyoura sand is shown in Figure 1. The soil's D_{50} is 0.26 mm, so if modeled as is, about 200 million particles are required with a time step size of 4×10^{-8} seconds,

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which will impose a significant computational burden to conduct the DEM simulation. To keep the computational cost manageable, linear scaling of the gradation is often adopted in the DEM research community. This study enlarges the original particle size by 30 times to obtain the revised size distribution as also shown in Figure 1. The size distribution is practically uniform, so particles are further approximated to spheres of 8 mm diameter. Consequently, about 6770 particles could be considered with a simulation time step size of 1×10^{-6} seconds.

The box is modeled with four vertical walls and a bottom plate. The particles are generated above the modeled box with the constraint of the aforestated grain size distribution and then poured into the box. The target void ratio of 0.76 is achieved through a trial and error procedure by adjusting the DEM modeling parameters including the coefficient of friction. Once the box is filled up, the pouring is stopped and any particles above the box are removed. The DEM modeling parameters are then reset to the desired values. The model footing is placed above the bed and pushed into the soil, from which the penetration force is measured against the penetration depth.

The DEM modeling parameters are calibrated with the experiment data obtained in $1g$ gravity condition, which are summarized in Table 1. Figure 2 shows the DEM simulation of penetration in $1g$ gravity condition. The Hertz contact model is considered with the history effect of tangential displacement. LIGGGHTS internally computes the contact properties including the contact stiffnesses from the material properties and the overlap between particles in contact (DCS Computing Gmbh 2016b).

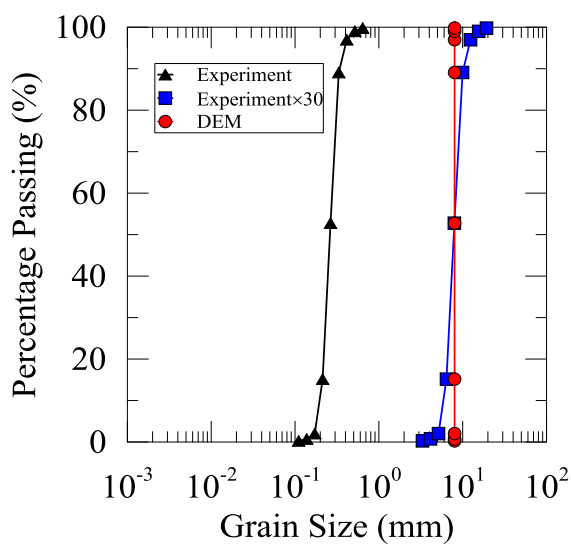

Figure 1. Grain size distribution of Toyoura sand

RESULTS AND DISCUSSIONS

With the calibrated DEM modeling parameters for $1g$ gravity condition, the DEM simulations are performed for two different reduced gravity conditions ($1/2g$, $1/6g$) to validate the soil behavior. The density of modeled particles is then scaled to $1/2\rho$ and $1/6\rho$ and used for the simulation of soil bearing resistance in Earth gravity (g). Lastly, the simulation results are compared to the results obtained for reduced gravity conditions.

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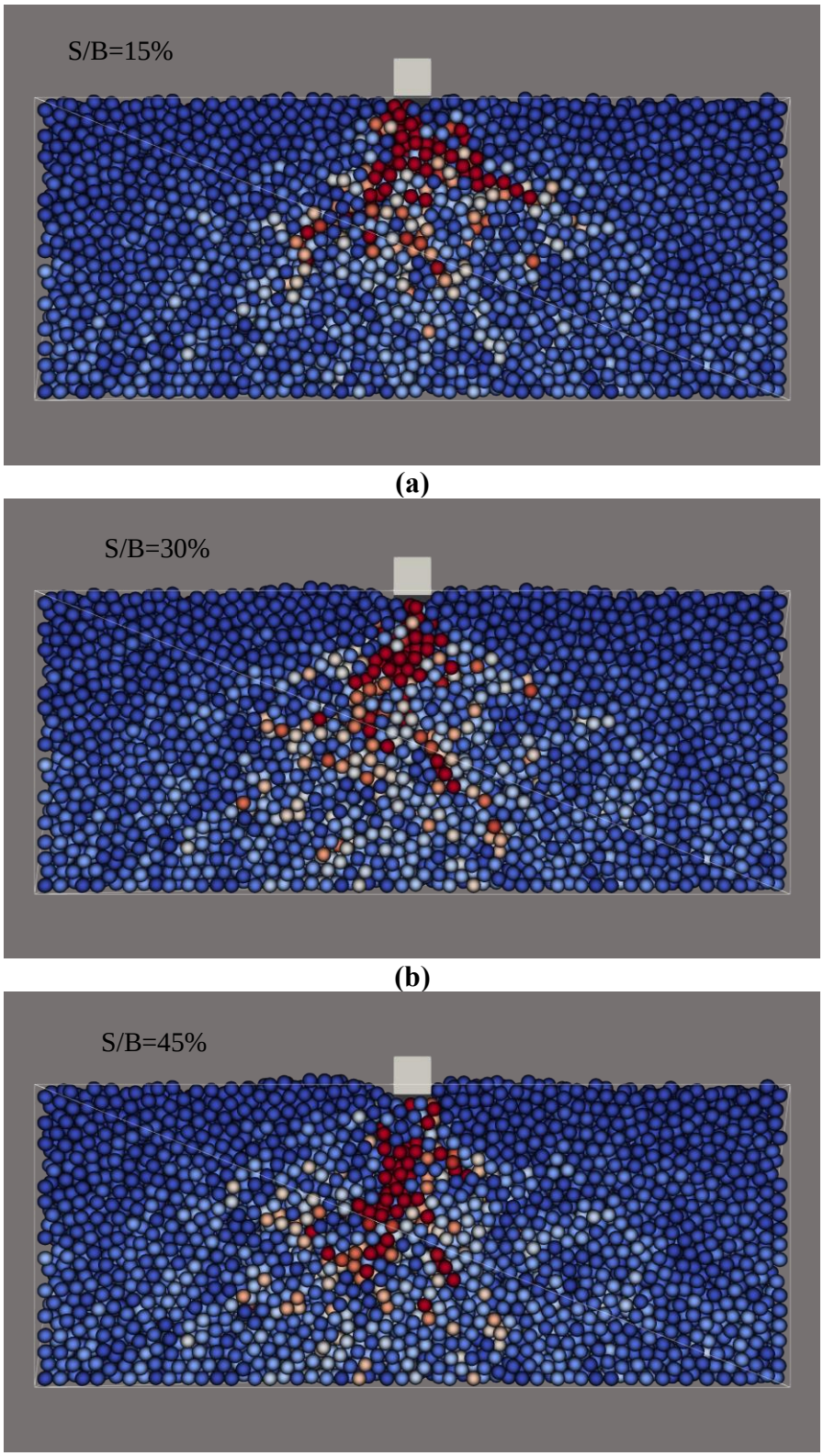

Figure 2. Contact force visualized with the penetration in 1*g* gravity, where S represents the settlement, and B is the footing width; (a) S/B = 15%, (b) 30%, and (c) 45%

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Table 1. DEM modeling parameters and values used for the simulations

DEM modeling parameter		Value
g	Gravity (g = 9.8 m/s^2)	1/6g, 1/2g, 1g for 1ρ particle density
ρ	Particle density (ρ = 2650 kg/m^3)	1/6ρ, 1/2ρ for 1g gravity condition
E	Young's modulus of particle	1×10^{10} Pa
ν	Poisson's Ratio	0.3
μ	Coefficient of friction	0.5 (~ tan 27°)
μ_r	Coefficient of rolling friction	0.3

Bearing Resistance of Soil with 1ρ Particle Density in 1g, 1/2g, 1/6g Gravities

Each simulation is repeated three times from which the average response is obtained to eliminate the effect of initial particle arrangements (Lee et al. 2012; Qian et al. 2018). The experiment data in 1g gravity is used to calibrate the DEM modeling parameters and is compared to the DEM simulation result as shown in Figure 3. The experiment data and simulation results for reduced gravity conditions (1/2g, 1/6g) are shown in Figure 4 and Figure 5. The particle density is maintained to 1ρ (2650 kg/m^3) in all simulations. Despite the limited fluctuations shown in simulation results due to the larger particle size adopted compared to the actual size, the simulation results are overall comparable to the experiment data.

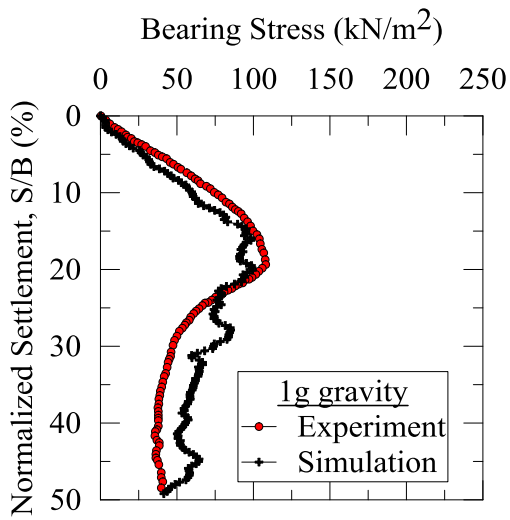

Figure 3. Experiment data and simulated bearing resistance of Toyoura sand using DEM in 1g gravity; Experiment data is digitized from Kobayashi et al. (2009)

Figure 6 compares the simulated bearing resistance of soil with 1/2ρ particle density in 1g gravity condition to the simulation result of soil with 1ρ particle density in 1/2g gravity (presented in Figure 4). Overall good agreement is shown between the simulation results which demonstrates Terzaghi's formula holds true for the developed hypothesis, i.e., the soil bearing resistance in a different gravity condition can be studied in the Earth gravity using synthetic soil particles whose density is proportionally scaled. Figure 7 also shows the bearing resistance of soil with 1/6ρ particle density in 1g gravity condition is comparable to the result with 1ρ particle density in 1/6g gravity.

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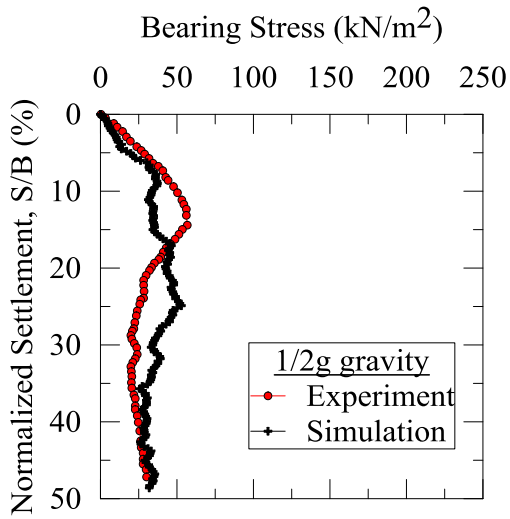

Figure 4. Experiment data and simulated bearing resistance of Toyoura sand using DEM in 1/2g gravity; Experiment data is digitized from Kobayashi et al. (2009)

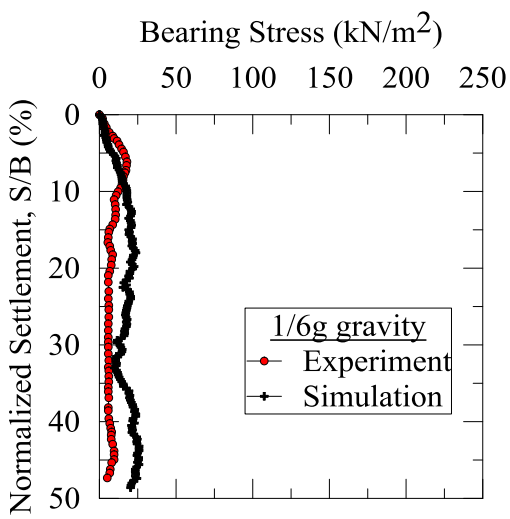

Figure 5. Experiment data and simulated bearing resistance of Toyoura sand using DEM in 1/6g gravity; Experiment data is digitized from Kobayashi et al. (2009)

Bearing Resistance of Soil with 1/2ρ and 1/6ρ Particle Densities in 1g Gravity

CONCLUDING REMARKS

This study leverages Terzaghi's bearing capacity theory to demonstrate that the gravity effect on the soil bearing resistance can be reproduced by adopting a synthetic soil with proportionally scaled particle density. Therefore, this approach is an alternative to existing experimental methods and will facilitate the understanding of extraterrestrial soil behavior in a general laboratory setting. With recent advances in the 3D printing technology, a large quantity of 3D-printed synthetic particles can be easily produced with controlled particle density. Therefore, the mechanical

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behavior of planetary soil can be simulated in an inexpensive manner. Consequently, this approach has a great potential to improve the space exploration-related education as well as the research.

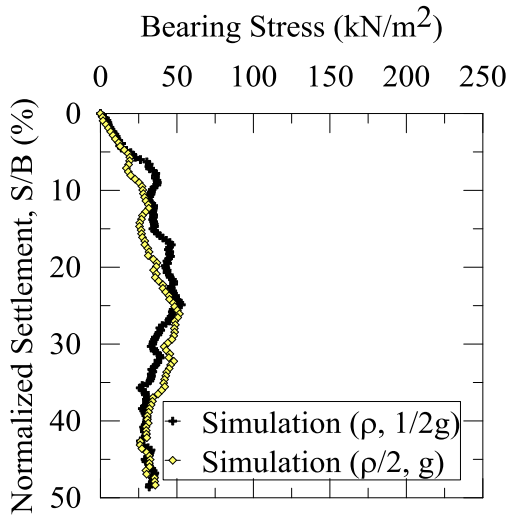

Figure 6. Simulated bearing resistance of synthetic sand with 1/2ρ particle density in 1g gravity condition

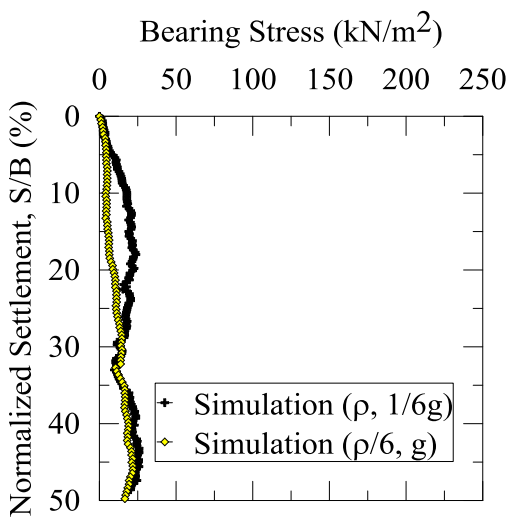

Figure 7. Simulated bearing resistance of synthetic sand with 1/6ρ particle density in 1g gravity condition

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